SCATS · Level 1 · Foundation

The traffic light that grades its own green time

How SCATS watches one loop per lane, measures how much of each green was really used, and adjusts the next cycle.

6 ideas, each building on the last

1 What SCATS is

60,000+ intersections reported to run SCATS worldwide

2 Two levels of control

±6 s largest normal cycle change per cycle (this lab)

3 The timing vocabulary

48 / 32 s greens of a 90 s cycle split 60/40

4 Measuring demand: the degree of saturation

DS 0.90 g 40 s, T 20 s, 10 cars, h 1.6 s

5 Fixed time, actuated and SCATS

−30% SCATS delay vs fixed time in Lab F3

6 Where fixed-time plans come from

h = 1.62 s calibrated standard space time (Lab F2)

Three surprises to start

SCATS measures empty road.

Instead of counting cars, SCATS adds up the moments when nothing is over the loop during green. Wasted green is easy to see; used green is whatever is left.

It runs on a huge scale.

SCATS was developed in Sydney from the 1970s and is owned today by Transport for NSW. It is reported to run more than 60,000 intersections in over 200 cities.

A truck is handled fairly without being recognised.

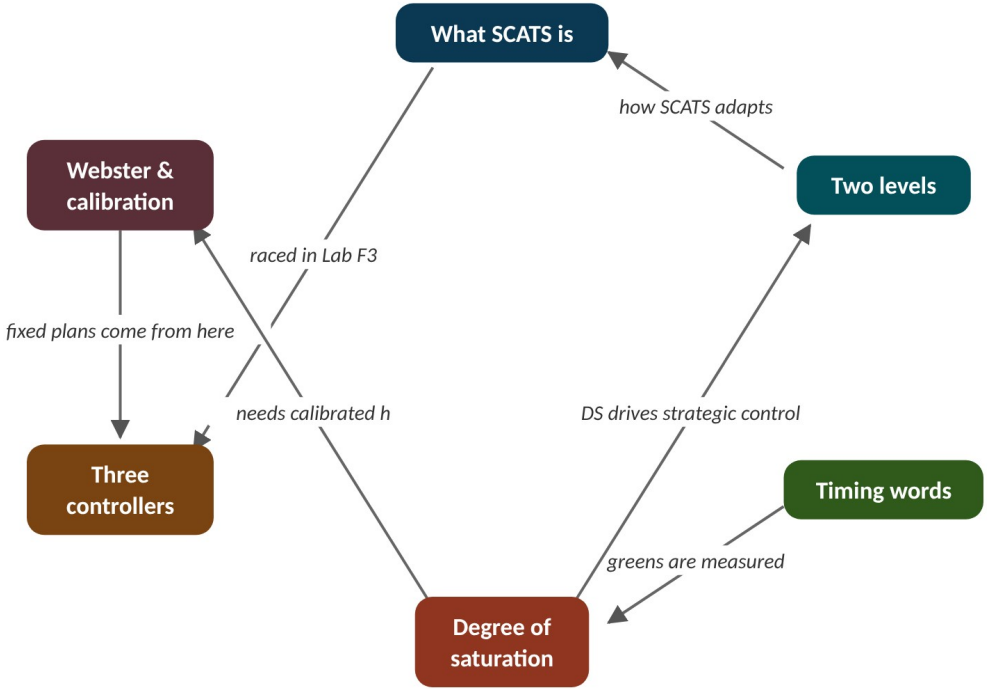
A truck covers the loop for longer, so it leaves less empty time, exactly as it uses more green. SCATS never needs to know it was a truck.

Words you will meet

Intersection	A place where roads meet and traffic lights take turns.
Phase	A set of movements that get green together, for example 'main road both ways' (phase A).
Cycle	One full turn through all phases, including the changeover times, for example 90 s.
Split	How the green time is shared between phases, for example 60% to A and 40% to B.
Intergreen	Amber plus all-red between two phases. Nobody moves much, so it is 'lost' time.

Offset	When one intersection's cycle starts compared with its neighbour, so a green wave can form.
Stop-line loop	A wire loop in the road just before the stop line that knows when a vehicle is over it.
Degree of saturation (DS)	How much of a green was really used: 1 means fully used, 0.4 means mostly empty road.
Subsystem	A small group of intersections that share one cycle length.

The map of today's ideas



1 What SCATS is

IN PLAIN WORDS

SCATS is like a teacher marking homework every day instead of once a year. Each day she sees what went wrong and adjusts tomorrow's lesson a little. Small daily fixes keep the class on track.

- Adaptive: timings change every cycle
- Coordinated: neighbours share a cycle
- Plan-based: chooses from approved plans
- Grades each green, nudges the next

60,000+

intersections reported to run SCATS worldwide

Worked example: How often does SCATS think?

One intersection running a 90 s cycle for one hour.

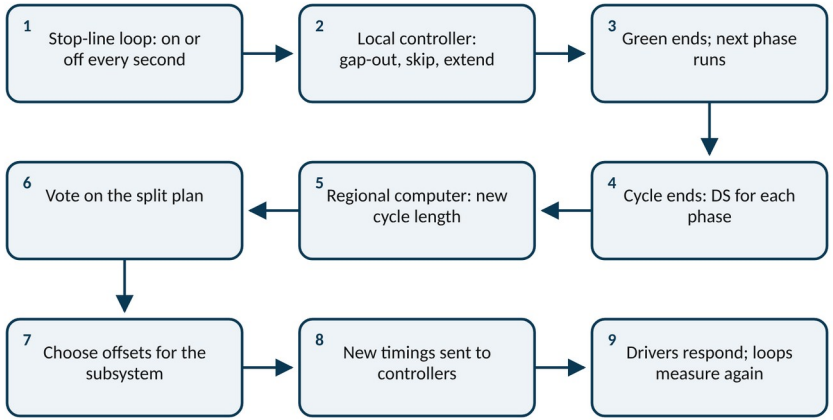
1. Cycles per hour = 3600 ÷ 90 = **40**.

2. Strategic decisions (cycle length, split, offset) happen once per cycle: **40** per hour.

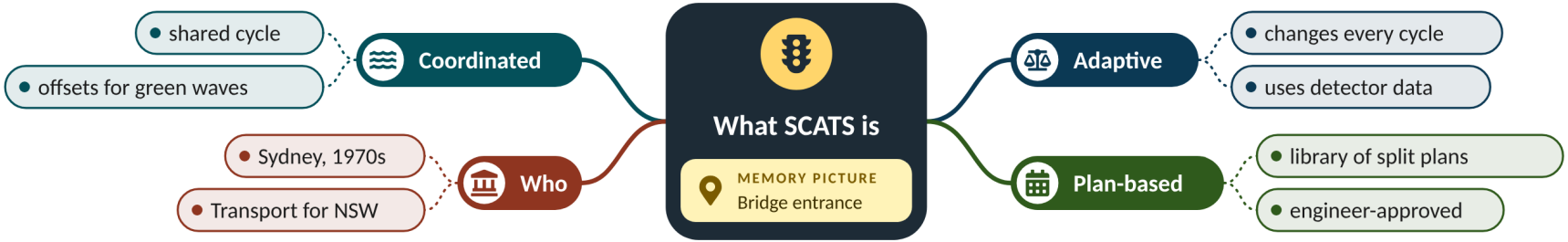
3. Tactical checks (should this green end now?) happen every second: 3600 per hour.

4. So for every strategic decision there are 3600 ÷ 40 = **90** tactical checks.

Forty careful decisions an hour, each backed by ninety quick checks. Each correction can be small because another one is coming within two minutes.



Remember it: What SCATS is



PICTURE THIS · STOP 1: BRIDGE ENTRANCE

At the start of the bridge a conductor in a huge coat waves a baton at the traffic lights. Instead of one score she flips between four thick music books, and after every bar she scribbles a red mark on the page she just played.

TRAP

Adaptive does not mean 'anything goes'. SCATS keeps the cycle between a minimum and a maximum, changes it by a few seconds at a time, and picks splits from a small library of approved plans.

2 Two levels of control

IN PLAIN WORDS

A restaurant has waiters and a manager. Waiters react every second: this table is empty, serve the next one. The manager looks at the whole evening and decides how many tables and staff tonight needs.

- Tactical: every second, in the controller
- Gap-out ends empty greens early
- Strategic: every cycle, regional computer
- Cycle steered towards DS 0.9

±6 s

largest normal cycle change per cycle (this lab)

Worked example: Two cycle-length decisions

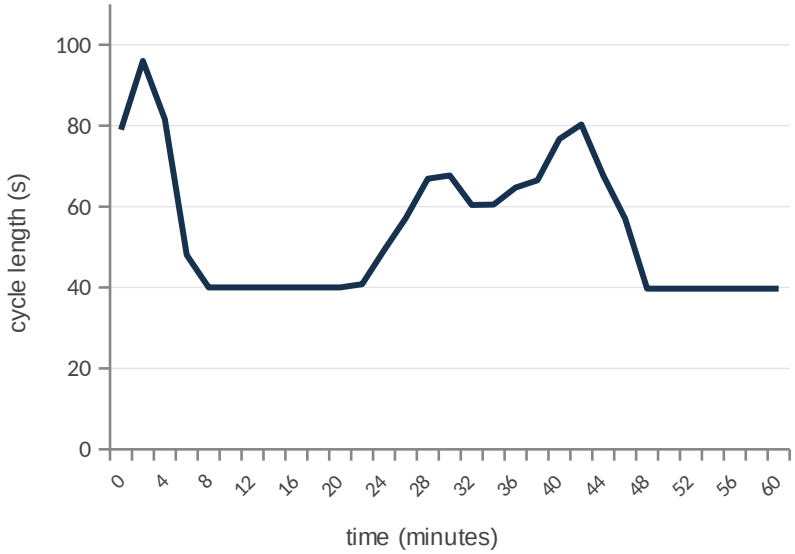
desired cycle = current cycle × DS ÷ 0.9

The cycle is 80 s. Case 1: the busiest phase had DS 0.99. Case 2: it had DS 0.60.

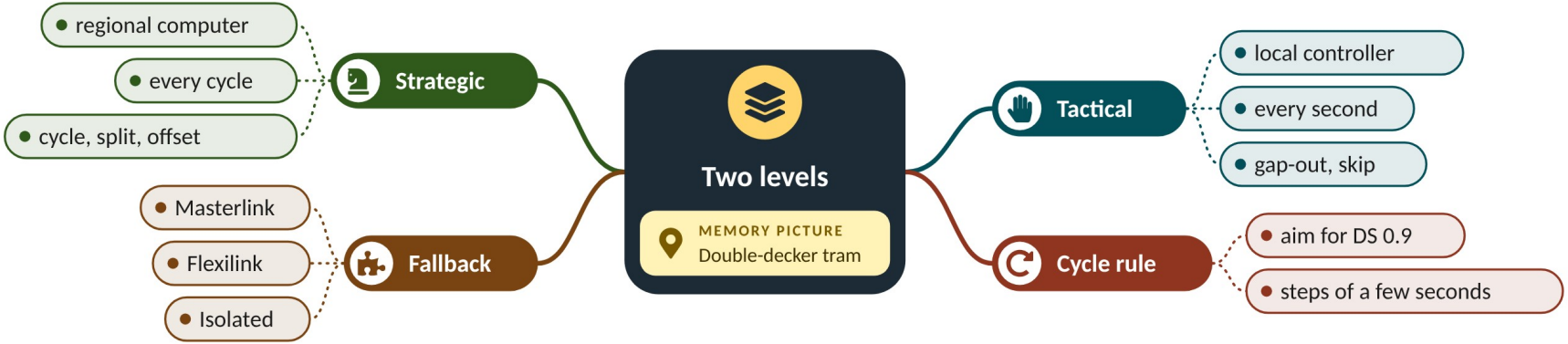
1. Case 1 desired: 80 × 0.99 ÷ 0.9 = **88 s**, which is 8 s longer.
2. DS is below 1, so the step limit is +6 s: new cycle = 80 + 6 = **86 s**.
3. Case 2 desired: 80 × 0.60 ÷ 0.9 = **53.3 s**, which is 26.7 s shorter.
4. The step limit is −6 s: new cycle = 80 − 6 = **74 s**.

SCATS moves towards the cycle that would give DS 0.9, but only a few seconds per cycle, so one odd cycle cannot swing the timings.

SCATS cycle length over one hour in Lab F3 (SUMO).



Remember it: Two levels



PICTURE THIS · STOP 2: DOUBLE-DECKER TRAM

Halfway across the bridge a double-decker tram rattles by. Downstairs a driver slaps a big button every single second, shouting 'empty, next!'. Upstairs a captain with a clipboard looks through a telescope and, once a lap, turns a dial by exactly six clicks.

TRAP

Gap-out is tactical, not strategic. Ending a green early because the road emptied happens in the local controller within seconds; it is not a decision of the regional computer.

3 The timing vocabulary

IN PLAIN WORDS

A cycle is a pizza. Each phase gets a slice; the split says how big each slice is. The crust between slices is the intergreen: it comes with every slice and nobody really eats it.

- Phase: movements that run together
- Cycle: all phases, intergreens included
- Split: how green is shared
- Intergreen: amber plus all-red, lost

48 / 32 s

greens of a 90 s cycle split 60/40

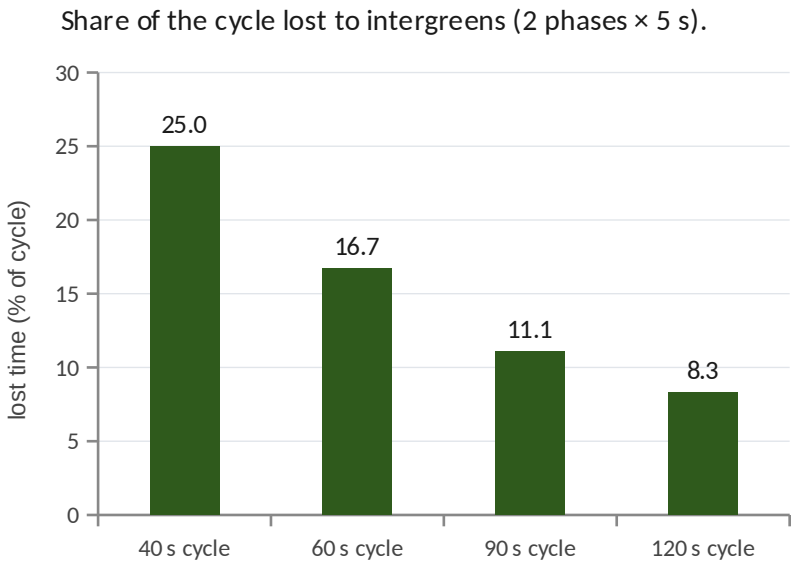
Worked example: Building a 90 s cycle

$$\text{total green} = \text{cycle} - (\text{number of phases} \times \text{intergreen})$$
$$\text{green}_i = \text{total green} \times \text{split share}_i$$

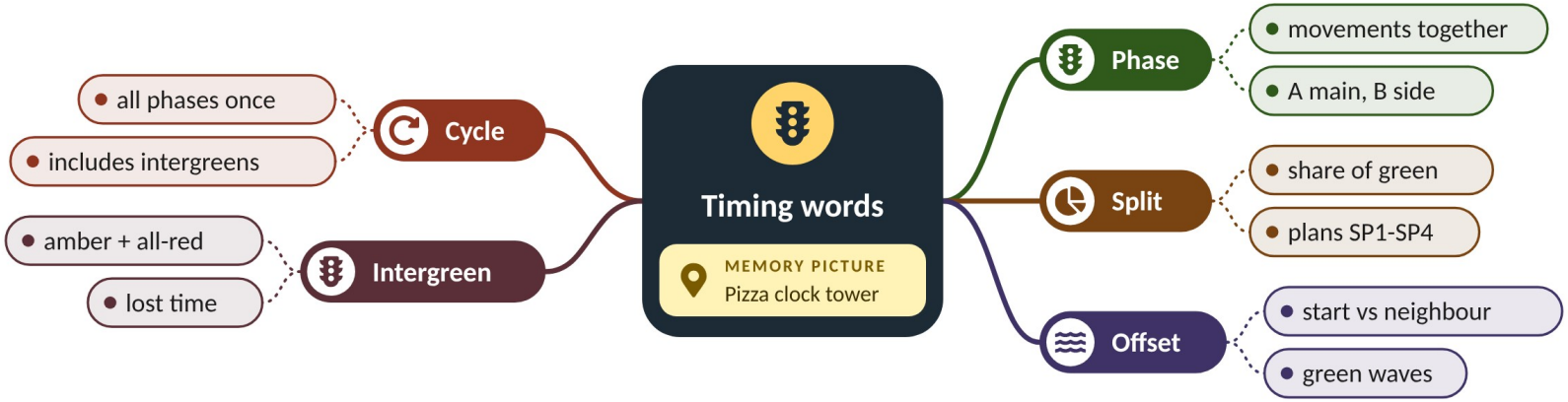
Two phases, 5 s intergreen each, split 60% to A and 40% to B.

1. Lost time = 2 × 5 = **10 s**.
2. Total green = 90 − 10 = **80 s**.
3. Phase A green = 80 × 0.60 = **48 s**; phase B green = 80 × 0.40 = **32 s**.
4. Share of the cycle lost = 10 ÷ 90 = **11.1%**. With a 60 s cycle it would be 10 ÷ 60 = **16.7%**.

Longer cycles waste a smaller share of time, but make drivers wait longer at red. Choosing the cycle is a balance.



Remember it: Timing words



PICTURE THIS · STOP 3: PIZZA CLOCK TOWER

On a tower above the bridge, the clock face is a giant pizza. The hand sweeps round once per cycle. Two slices, one big and one small, are painted A and B, and a thick burnt crust between them crunches loudly every time the hand passes.

TRAP

The cycle includes the intergreens. A 90 s cycle with two 5 s intergreens has only 80 s of green to share, not 90.

4 Measuring demand: the degree of saturation

IN PLAIN WORDS

Think of a supermarket checkout. Even when the queue is endless there is a short pause between customers. Any pause longer than that normal pause is the cashier waiting for nobody: wasted time.

- Loop measures gap time during green
- Wasted green = T − n·h
- DS = used green ÷ green
- VK turns DS into car-equivalents

DS 0.90

g 40 s, T 20 s, 10 cars, h 1.6 s

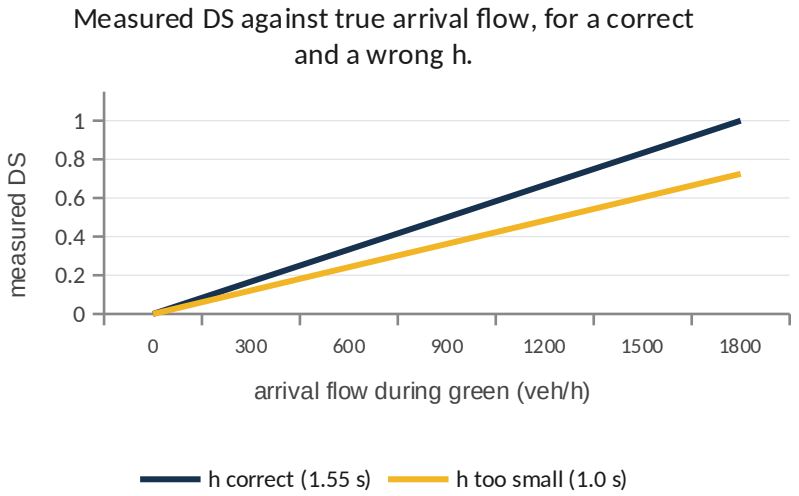
Worked example: DS and VK from one green

$$g' = g - (T - n \cdot h)$$
$$DS = g' \div g$$
$$VK = DS \times g \times MF \div 3600$$

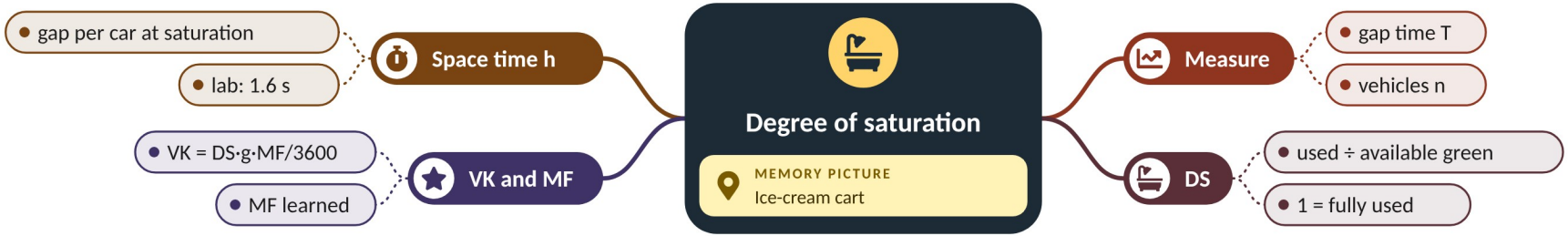
Green g = 40 s. The loop was empty for T = 20 s. n = 10 vehicles crossed. h = 1.6 s, MF = 1800 veh/h.

1. Expected gaps: n × h = 10 × 1.6 = **16 s**.
2. Wasted green: T − n·h = 20 − 16 = **4 s**.
3. Used green: g′ = 40 − 4 = **36 s**.
4. DS = 36 ÷ 40 = **0.90**.
5. VK = 0.90 × 40 × 1800 ÷ 3600 = **18** car-equivalents.

90% of the green was used: right at the usual target. If h were set wrongly, DS would be wrong too, which is why h is calibrated.



Remember it: Degree of saturation



PICTURE THIS · STOP 4: ICE-CREAM CART

By the bridge's far pylon an ice-cream seller serves a queue. Between customers she holds up a ruler 1.6 long; any gap longer than the ruler, she drops a cone in a bin labelled WASTED. A scoreboard shows 'used ÷ open time'.

TRAP

DS is not 'number of cars ÷ capacity' counted in vehicles. It is measured in time from the loop's gaps, and it depends on h. A wrong h gives a wrong DS for every cycle.

5 Fixed time, actuated and SCATS

IN PLAIN WORDS

Fixed time is a lift that stops at every floor whether anyone is waiting or not. Actuated is a lift that skips empty floors. SCATS also skips empty floors, and every few minutes rethinks its whole timetable from how busy it was.

- Fixed: one plan, ignores traffic
- Actuated: ends empty greens early
- SCATS: also re-plans every cycle
- Lab F3: 20.4 vs 15.0 vs 14.3 s

−30%

SCATS delay vs fixed time in Lab F3

Worked example: How big are the differences?

improvement = (old − new) ÷ old × 100%

Lab F3 mean delays: fixed 20.44 s, actuated 14.96 s, SCATS 14.27 s.

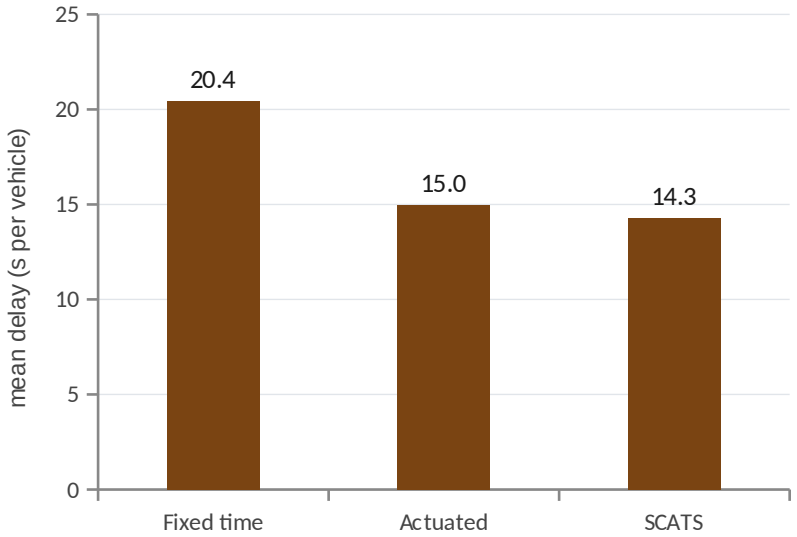
1. SCATS vs fixed: (20.44 − 14.27) ÷ 20.44 = 6.17 ÷ 20.44 = **30.2%** less delay.

2. SCATS vs actuated: (14.96 − 14.27) ÷ 14.96 = 0.69 ÷ 14.96 = **4.6%** less delay.

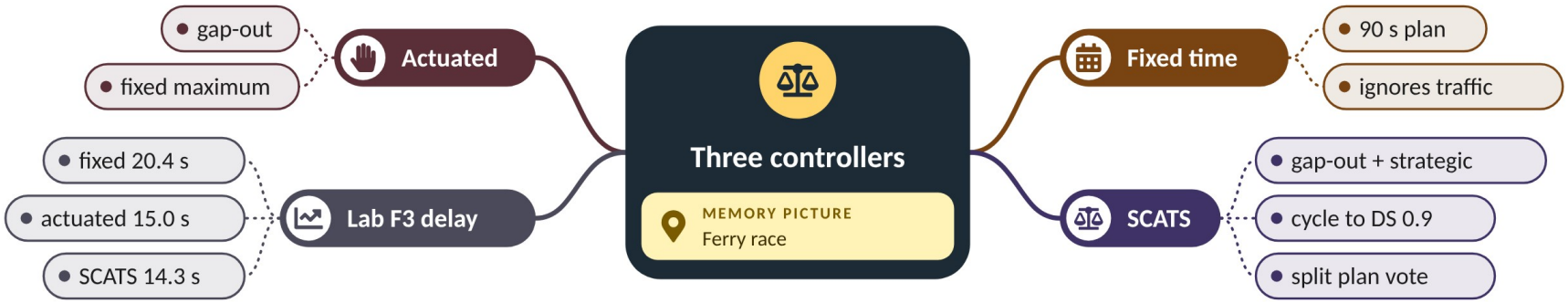
3. Actuated vs fixed: (20.44 − 14.96) ÷ 20.44 = **26.8%** less delay.

Most of the gain over fixed time comes from reacting to gaps; SCATS's strategic layer adds a little more at a single intersection.

Mean delay per vehicle in Lab F3 (SUMO, one intersection, one hour).



Remember it: Three controllers



PICTURE THIS · STOP 5: FERRY RACE

Below the bridge three ferries race to the quay. A rusty clockwork ferry chugs in at twenty and a half. A jumpy speedboat arrives at fifteen, and just ahead of it, at fourteen and a quarter, glides a ferry whose captain studies a chart after every lap.

TRAP

Do not conclude 'SCATS is only 5% better than actuation'. This test is one isolated intersection. Coordination across a corridor, which actuation cannot do, is where the bigger gains appear.

6 Where fixed-time plans come from

IN PLAIN WORDS

Webster is a tailor who measures you once and sews a perfect suit. It fits beautifully on that day. If you gain weight or wear a coat underneath, the suit no longer fits, and nobody re-measures you.

- Webster designs one fixed plan
- $C0 = (1.5L + 5) \div (1 - Y)$
- Cycle explodes as Y nears 1
- SCATS needs calibrated h and MF

h = 1.62 s

calibrated standard space time (Lab F2)

Worked example: Webster for a two-phase intersection

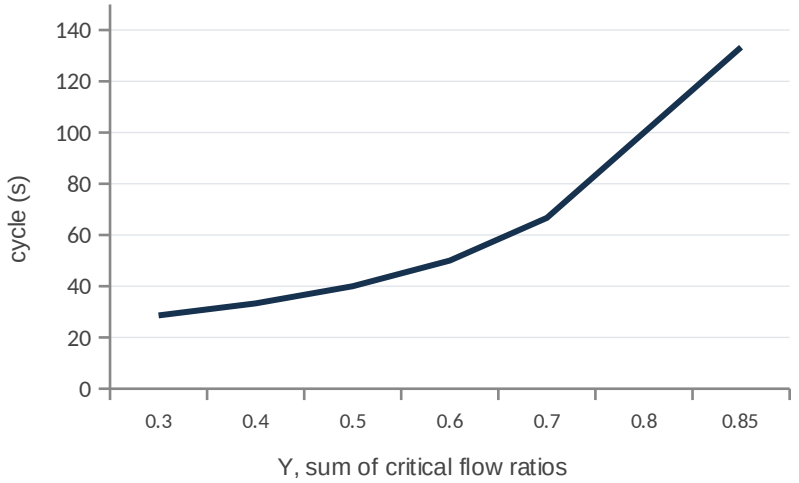
$$C_0 = (1.5\,L + 5) \div (1 - Y)$$
$$g_i = (C_0 - L) \times y_i \div Y$$

Lost time L = 10 s. Main road y_A = 0.40, side road y_B = 0.20.

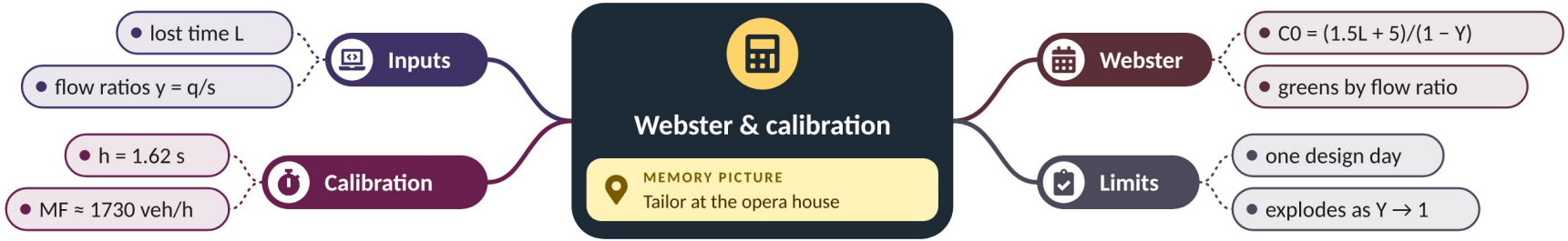
1. Y = 0.40 + 0.20 = **0.60**.
2. C_0 = (1.5 × 10 + 5) ÷ (1 − 0.60) = 20 ÷ 0.40 = **50 s**.
3. Green to share = 50 − 10 = **40 s**.
4. Main road: 40 × 0.40 ÷ 0.60 = **26.7 s**. Side road: 40 × 0.20 ÷ 0.60 = **13.3 s**.

A good fixed plan for exactly these flows: 50 s cycle, 26.7 s and 13.3 s of effective green.

Webster's optimum cycle as demand grows (L = 10 s).



Remember it: Webster & calibration



PICTURE THIS · STOP 6: TAILOR AT THE OPERA HOUSE

On the opera house steps a tailor stitches a suit onto a dancer. His tape measure is printed '1.5 L + 5', and the suit fits perfectly until the dancer eats a giant pie and the buttons fly off. Next door an assistant re-measures a second dancer after every pirouette, holding a ruler marked 1.62.

TRAP

If Y ≥ 1 no cycle length works: demand exceeds capacity. A huge cycle from Webster is a warning that the intersection is near capacity, not a plan to use.

Three ways to run a traffic light

	Fixed time	Actuated	SCATS
Detectors	none needed	stop-line loops	stop-line loops in every lane
Ends a green early?	no	yes, by gap-out	yes, by gap-out
Changes the cycle?	no	no (fixed maximum)	every cycle, towards DS 0.9
Coordinates neighbours?	if designed so	no	yes: subsystems, offsets
Lab F3 delay	20.44 s	14.96 s	14.27 s

Things beginners mix up

This…	…is not this	The fact that tells them apart
Tactical	**Strategic**	Tactical decides every second in the local controller; strategic decides once per cycle in the regional computer.
Cycle	**Total green**	The cycle includes the intergreens; total green is what is left after them.
Gap time T	**Wasted green**	Wasted green is T minus the normal gaps n·h.
DS	**Vehicle count ÷ capacity**	DS is measured in time from loop gaps, so trucks count for more automatically.
Adaptive	**Unlimited**	SCATS changes timings in small steps and from approved plans.

Check yourself (1 of 2)

A 90 s cycle has two phases with 5 s intergreen each, split 70/30. What is phase A's green?

A. 63 s

B. 56 s

C. 59.5 s

D. 45 s

The cycle is 100 s and the busiest phase had DS 0.72. Using this lab's rule, what is the next cycle?

A. 80 s

B. 94 s

C. 106 s

D. 100 s

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

In Lab F3, which controller had the lowest mean delay?

A. Fixed time

B. Actuated

C. SCATS

D. All equal

A truck and a car pass the loop during a saturated green. How does DS treat the truck?

A. Ignores it

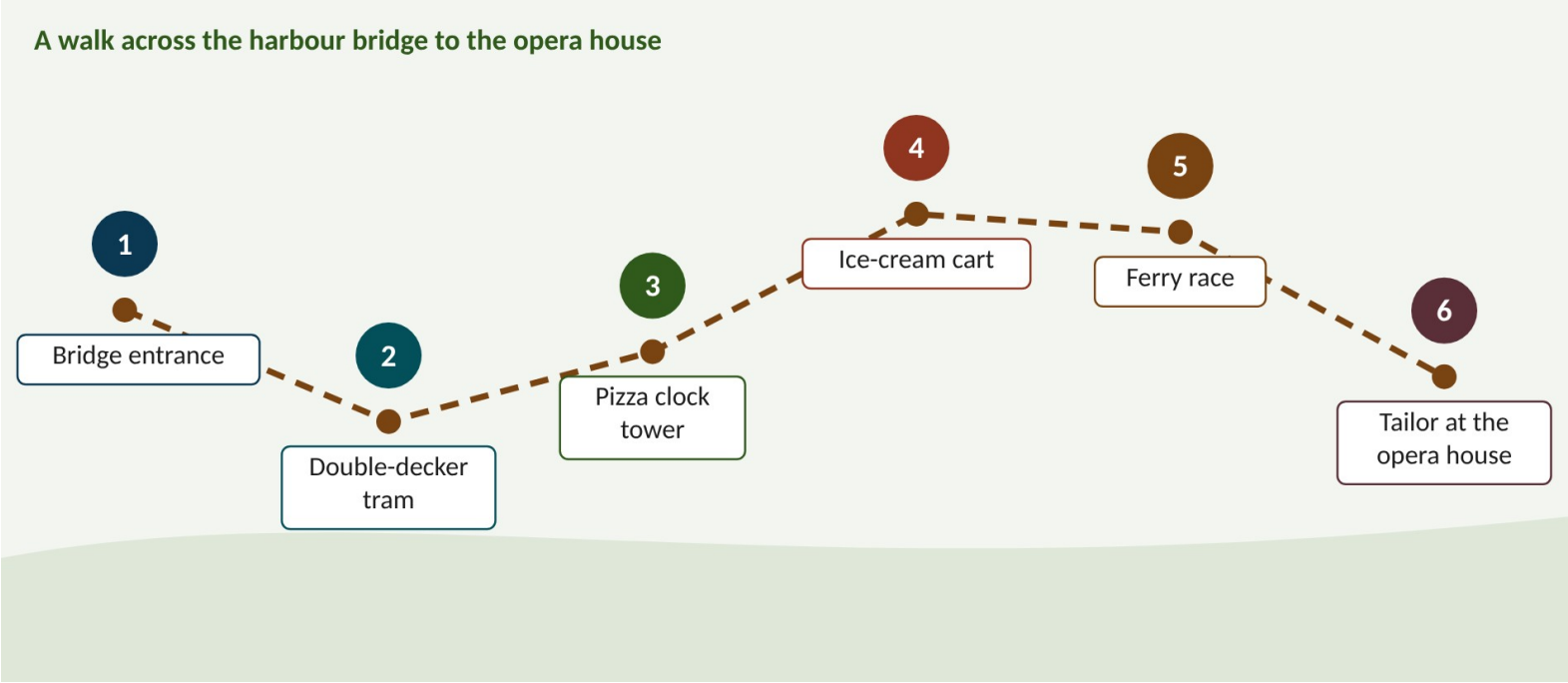
B. Counts it as one car

C. It leaves less empty time, so it raises DS as it uses more green

D. It lowers DS

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Bridge entrance = What SCATS is · 2 Double-decker tram = Two levels · 3 Pizza clock tower = Timing words · 4 Ice-cream cart = Degree of saturation · 5 Ferry race = Three controllers · 6 Tailor at the opera house = Webster & calibration

Numbers to remember

- Target DS ≈ 0.9
- Cycle 40-140 s, step +6 (+9 if DS ≥ 1) / −6 s
- Gap-out setting 3 s; intergreen 5 s
- h = 1.62 s, MF ≈ 1730 veh/h (Lab F2)
- g 40, T 20, n 10, h 1.6 → DS 0.90, VK 18
- Lab F3 delay: 20.44 / 14.96 / 14.27 s
- Webster: L 10, Y 0.6 → C_0 = 50 s

Keep going

Study guide: SCATS_Foundation_Study_Guide.pdf · Interactive labs and SUMO code: smrahmanbd.github.io/scats-learning-lab